Mengjie Yang

Ph.D. Candidate in Physics, National University of Singapore
Expected Graduation: March 2027
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Research Interests

Topological physics; Open quantum systems; Non-equilibrium quantum many-body dynamics; Quantum measurement and feedback; Casimir and quantum fluctuation physics.

Education

National University of Singapore Singapore
Ph.D. Candidate in Physics *Aug 2023–Present*

- Supervisor: Prof. Ching Hua Lee.
- Thesis topic: non-Hermitian topology, open-system dynamics, and boundary-sensitive spectral phenomena.

Shanghai Jiao Tong University Shanghai, China
B.Sc. in Physics, Zhiyuan Honors Program (top 5%) *Sep 2019–Jun 2023*

- Supervisor: Prof. Luqi Yuan.
- Bachelor thesis on subradiant modes in chiral-waveguide-coupled atomic arrays; awarded Best Bachelor Thesis (top 1%).

Publications

1. M. Yang and C. H. Lee, "Beyond the Non-Hermitian Skin Effect: Scaling-Controlled Topology from Exceptional-Bound Bands," *Advanced Science*, e2523989 (2026).
2. M. Yang and C. H. Lee, "Reversing non-Hermitian skin accumulation with a non-local transverse switch," *Communications Physics* (2026).
3. M. Yang, L. Yuan, and C. H. Lee, "Non-Hermitian strong bosonic clustering through interaction-induced caging," *Communications Physics* **8**, 388 (2025).
4. M. Yang and C. H. Lee, "Percolation-Induced $\mathcal{PT}$ Symmetry Breaking," *Physical Review Letters* **133**, 136602 (2024).
5. M. Yang, L. Wang, X. Wu, H. Xiao, D. Yu, L. Yuan, and X. Chen, "Concentrated subradiant modes in a one-dimensional atomic array coupled with chiral waveguides," *Physical Review A* **106**, 043717 (2022).
6. M. Yang and C. H. Lee, "Beyond symmetry protection: Robust feedback-enforced edge states in non-Hermitian stacked quantum spin Hall systems," arXiv:2507.17295 (2025).

Ongoing Projects

† *denotes equal contribution.*

1. M. Yang, Alexander. Poddubny (Weizmann Institute), and C. H. Lee, "From logarithmic to scale-covariant criticality in nonlocal skin ladders," manuscript in preparation.
2. M. Yang and C. H. Lee, and A. Poddubny, "Activated mode theory for generalized real-complex transitions," manuscript in preparation.
3. M. Yang, S Rahul (PES University), and Giandomenico Palumbo (University of Coimbra), a study on topological metal, manuscript in preparation.
4. M. Yang and C. H. Lee, "Reflection-operator theory of non-Hermitian Casimir interactions: imaginary-axis regularity and exceptional-point scaling," manuscript in preparation.

5. T. Wang†, M. Yang†, and Flore Kunst (Max Planck Institute), "Exact solutions of nonreciprocal Su-Schrieffer-Heeger model with domain walls," manuscript in preparation.
6. K. Gao, J. Umar, M. Yang, C. H. Lee, a co-supervised Moire physics manuscript in preparation.
7. M. Yang, C. H. Lee, A. Poddubny, two-boson problems in a chiral Bose-Hubbard model, ongoing.
8. A co-authored review article on quantum computing with collaborators in Lee's research group

Invited and Contributed Presentations

1. *Max Planck Institute for the Science of Light*, Erlangen, Germany, Dec 2025. "Topological-Guided Gain and Feedback-Enforced Edge Dynamics in Non-Hermitian Systems."
2. *BLINH 2025*, Erice, Sicily, Italy, Dec 2025. "Percolation-induced PT-symmetry breaking."
3. *Quantum Connections*, Stockholm, Sweden, Jun 2025. "Percolation-induced PT-symmetry breaking"; "Non-Hermitian strong bosonic clustering through interaction-induced caging."
4. *Shanghai Jiao Tong University*, China, 2025. Invited seminar.
5. *IPS Meeting*, 2025. Invited talk on exceptional-bound bands; poster on non-Hermitian ultra-strong bosonic clustering.
6. *IPS Meeting*, 2024. Contributed talk on percolation-induced PT-symmetry breaking.

Honors and Awards

1. Faculty of Science PhD Conference Award, National University of Singapore, Nov 2024.
2. Outstanding Undergraduate Graduate, Shanghai Jiao Tong University, Jun 2023.
3. Best Bachelor Thesis (top 1%), Shanghai Jiao Tong University, Jun 2023.
4. Zhiyuan Honors Scholarship, Shanghai Jiao Tong University, awarded four times, 2019–2023.

Academic Service

1. Peer reviewer, *Communications Physics* (3 reviews, 2024–2025).
2. Peer reviewer, *Nature Communications* (1 review, 2026).

Teaching Experience

1. Teaching Assistant, *Advanced Solid State Physics*, National University of Singapore, Semester 1, 2024, 2025, and 2026.
2. Teaching Assistant, *Computational Thinking for Scientists*, National University of Singapore, Semester 2, 2025.